例 1 设$X(t)$是均值为零、自相关函数为$R_X(\tau)$平稳过程。令$Z(t)=X(t)+\mathrm{j}\cdot\hat{X}(t)$，其中$\hat{X}(t)$是$X(t)$的希尔伯特变换。求$Z(t)$的均值$\mathrm{E}[Z(t)]$、自相关函数$\mathrm{E}[Z(t+\tau)Z^*(t)]$、共轭相关函数$\mathrm{E}[Z(t+\tau)Z(t)]$。

解一：零均值平稳过程的希尔伯特变换仍然是零均值平稳过程，故$\hat{X}(t)$均值为零，因此$\mathrm{E}[Z(t)]=\mathrm{E}\left[X(t)+\mathrm{j}\cdot\hat{X}(t)\right]=0$。

$Z(t)$的自相关函数是

$$\begin{aligned}\mathrm{E}\left[Z(t+\tau)Z^*(t)\right]&=\mathrm{E}\left[\left\{X(t+\tau)+\mathrm{j}\cdot\hat{X}(t+\tau)\right\}\cdot\left\{X(t)-\mathrm{j}\cdot\hat{X}(t)\right\}\right]\\&=\mathrm{E}\left[X(t+\tau)X(t)\right]-\mathrm{j}\cdot\mathrm{E}\left[X(t+\tau)\hat{X}(t)\right]\\&\quad+\mathrm{j}\cdot\mathrm{E}\left[\hat{X}(t+\tau)X(t)\right]-\mathrm{j}^2\cdot\mathrm{E}\left[\hat{X}(t+\tau)\hat{X}(t)\right]\\&=R_X(\tau)-\mathrm{j}\cdot R_{X\hat{X}}(\tau)+\mathrm{j}\cdot R_{\hat{X}X}(\tau)+R_{\hat{X}}(\tau)\end{aligned}$$

根据零均值平稳过程希尔伯特变换的性质，$R_{\hat{X}}(\tau)=R_X(\tau)$，$R_{\hat{X}X}(\tau)=\hat{R}_X(\tau)$，$R_{X\hat{X}}(\tau)=R_{\hat{X}X}(-\tau)=\hat{R}_X(-\tau)$。$R_X(\tau)$是偶函数，其希尔伯特变换$\hat{R}_X(\tau)$是奇函数。故$R_{X\hat{X}}(\tau)=\hat{R}_X(-\tau)=-\hat{R}_X(\tau)$。因此

$$\mathrm{E}\left[Z(t+\tau)Z^*(t)\right]=2R_X(\tau)+\mathrm{j}\cdot2\hat{R}_X(\tau)$$

$Z(t)$的共轭相关函数是

$$\begin{aligned}\mathrm{E}\left[Z(t+\tau)Z(t)\right]&=\mathrm{E}\left[\left\{X(t+\tau)+\mathrm{j}\cdot\hat{X}(t+\tau)\right\}\cdot\left\{X(t)+\mathrm{j}\cdot\hat{X}(t)\right\}\right]\\&=\mathrm{E}\left[X(t+\tau)X(t)\right]+\mathrm{j}\cdot\mathrm{E}\left[X(t+\tau)\hat{X}(t)\right]\\&\quad+\mathrm{j}\cdot\mathrm{E}\left[\hat{X}(t+\tau)X(t)\right]+\mathrm{j}^2\cdot\mathrm{E}\left[\hat{X}(t+\tau)\hat{X}(t)\right]\\&=R_X(\tau)+\mathrm{j}\cdot R_{X\hat{X}}(\tau)+\mathrm{j}\cdot R_{\hat{X}X}(\tau)-R_{\hat{X}}(\tau)\end{aligned}$$

代入$R_{\hat{X}}(\tau)=R_X(\tau)$，$R_{\hat{X}X}(\tau)=\hat{R}_X(\tau)$，$R_{X\hat{X}}(\tau)=\hat{R}_X(-\tau)=-\hat{R}_X(\tau)$，得到

$$\mathrm{E}\left[Z(t+\tau)Z(t)\right]=0$$

解二：零均值平稳过程通过滤波器后还是零均值平稳过程，故$\hat{X}(t)$均值为零，因此$\mathrm{E}[Z(t)]=\mathrm{E}\left[X(t)+\mathrm{j}\cdot\hat{X}(t)\right]=0$。

关系式$Z(t)=X(t)+\mathrm{j}\cdot\hat{X}(t)$给定了一个输入为$X(t)$、输出为$Z(t)$线性时不变系统，其冲激响应是$h(t)=\delta(t)+\mathrm{j}\cdot\frac{1}{\pi t}$。$Z(t)$是$X(t)$与$h(t)$的卷积，其自相关函数是

$$
\begin{aligned}
\mathrm{E}\left[Z(t+\tau)Z^{*}(t)\right] &= \mathrm{E}\left[\int_{-\infty}^{\infty}h(u)X(t+\tau-u)\mathrm{d}u\cdot\int_{-\infty}^{\infty}h^{*}(v)X(t-v)\mathrm{d}v\right] \\
&= \mathrm{E}\left[\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}h(u)h^{*}(v)X(t+\tau-u)X(t-v)\mathrm{d}u\mathrm{d}v\right] \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}h(u)h^{*}(v)R_X(\tau-u+v)\mathrm{d}u\mathrm{d}v \\
&= \int_{-\infty}^{\infty}h(u)\left\{\int_{-\infty}^{\infty}R_X(\tau-u+v)\left[\delta(v)-\frac{\mathrm{j}}{\pi v}\right]\mathrm{d}v\right\}\mathrm{d}u \\
&= \int_{-\infty}^{\infty}h(u)\left\{\int_{-\infty}^{\infty}R_X(\tau-u+v)\delta(v)\mathrm{d}v+\mathrm{j}\int_{-\infty}^{\infty}R_X(\tau-u-w)\frac{1}{\pi w}\mathrm{d}w\right\}\mathrm{d}u
\end{aligned}
\tag{1}
$$

花括号中第一项积分可利用狄拉克冲激的采样性，第二项积分是希尔伯特变换，于是

$$
\begin{aligned}
&\mathrm{E}\left[Z(t+\tau)Z^{*}(t)\right] = \int_{-\infty}^{\infty}\left[\delta(u)+\frac{\mathrm{j}}{\pi u}\right]\left\{R_X(\tau-u)+\mathrm{j}\cdot\hat{R}_X(\tau-u)\right\}\mathrm{d}u \\
&= \int_{-\infty}^{\infty}\delta(u)\left\{R_X(\tau-u)+\mathrm{j}\cdot\hat{R}_X(\tau-u)\right\}\mathrm{d}u+\mathrm{j}\int_{-\infty}^{\infty}\frac{1}{\pi u}\left\{R_X(\tau-u)+\mathrm{j}\cdot\hat{R}_X(\tau-u)\right\}\mathrm{d}u \\
&= R_X(\tau)+\mathrm{j}\cdot\hat{R}_X(\tau)+\mathrm{j}\left[\hat{R}_X(\tau)+\mathrm{j}\cdot\hat{\hat{R}}_X(\tau)\right]
\end{aligned}
\tag{2}
$$

注意希尔伯特变换等于实信号相移 90°（即正频率后移 90°，负频率前移 90°），两次希尔伯特变换是相移 180°，即$\hat{\hat{R}}_X(\tau)=-R_X(\tau)$，于是

$$
\mathrm{E}\left[Z(t+\tau)Z^{*}(t)\right] = 2R_X(\tau)+2\mathrm{j}\cdot\hat{R}_X(\tau) \tag{3}
$$

共轭相关函数是

$$
\begin{aligned}
\mathrm{E}\left[Z(t+\tau)Z(t)\right] &= \mathrm{E}\left[\int_{-\infty}^{\infty}h(u)X(t+\tau-u)\mathrm{d}u\cdot\int_{-\infty}^{\infty}h(v)X(t-v)\mathrm{d}v\right] \\
&= \mathrm{E}\left[\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}h(u)h(v)X(t+\tau-u)X(t-v)\mathrm{d}u\mathrm{d}v\right] \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}h(u)h(v)R_X(\tau-u+v)\mathrm{d}u\mathrm{d}v \\
&= \int_{-\infty}^{\infty}h(u)\left\{\int_{-\infty}^{\infty}R_X(\tau-u+v)\left[\delta(v)+\frac{\mathrm{j}}{\pi v}\right]\mathrm{d}v\right\}\mathrm{d}u \\
&= \int_{-\infty}^{\infty}\left[\delta(u)+\frac{\mathrm{j}}{\pi u}\right]\left\{R_X(\tau-u)-\mathrm{j}\cdot\hat{R}_X(\tau-u)\right\}\mathrm{d}u \\
&= R_X(\tau)-\mathrm{j}\cdot\hat{R}_X(\tau)+\mathrm{j}\cdot\left[\hat{R}_X(\tau)-\mathrm{j}\cdot\hat{\hat{R}}_X(\tau)\right]=0
\end{aligned}
\tag{4}
$$

以上结果说明$Z(t)=X(t)+\mathrm{j}\cdot\hat{X}(t)$是复平稳过程。

例 2 设$X(t)$是均值为零、自相关函数为$R_X(\tau)$平稳过程，将$X(t)$分别通过传递函数为$H_1(f)=1+\mathrm{sgn}(f)$、$H_2(f)=1-\mathrm{sgn}(f)$的线性时不变系统得到输出$Z_1(t),Z_2(t)$。求$Z_1(t),Z_2(t)$的互相关函数。

解：令$h_1(t),h_2(t)$分别代表$H_1(f),H_2(f)$对应的冲激响应。注意$H_1(f),H_2(f)$不满足共轭对称性，所以$h_1(t),h_2(t)$是复信号。$Z_1(t),Z_2(t)$是$X(t)$分别与$h_1(t),h_2(t)$的卷积。$Z_1(t),Z_2(t)$的互相关函数是

$$\begin{aligned}\mathrm{E}\left[Z_1(t+\tau)Z_2^*(t)\right]&=\mathrm{E}\left[\int_{-\infty}^{\infty}h_1(u)X(t+\tau-u)\mathrm{d}u\cdot\int_{-\infty}^{\infty}h_2^*(v)X(t-v)\mathrm{d}v\right]\\&=\mathrm{E}\left[\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}h_1(u)h_2^*(v)X(t+\tau-u)X(t-v)\mathrm{d}u\mathrm{d}v\right]\\&=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}h_1(u)h_2^*(v)R_X(\tau-u+v)\mathrm{d}u\mathrm{d}v\\&=\int_{-\infty}^{\infty}h_1(u)\left\{\int_{-\infty}^{\infty}R_X(\tau-u+v)h_2^*(v)\mathrm{d}v\right\}\mathrm{d}u\end{aligned}\tag{5}$$

将花括号中的内积转到频域：

$$\int_{-\infty}^{\infty}R_X(\tau-u+v)h_2^*(v)\mathrm{d}v=\int_{-\infty}^{\infty}\mathcal{F}\left\{R_X(\tau-u+v)\right\}\cdot\left[\mathcal{F}\left\{h(v)\right\}\right]^*\mathrm{d}v$$

式右的傅氏变换是对 v 进行。$R_X(v+\tau-u)$对v做傅氏变换结果是$P_X(f)\mathrm{e}^{\mathrm{j}2\pi f(\tau-u)}$，$h_2(t)$的傅氏变换是$H_2(f)=1-\mathrm{sgn}(f)$，于是

$$\begin{aligned}\int_{-\infty}^{\infty}R_X(\tau-u+v)h_2^*(v)\mathrm{d}v&=\int_{-\infty}^{\infty}P_X(f)\mathrm{e}^{\mathrm{j}2\pi f(\tau-u)}\left[1-\mathrm{sgn}(f)\right]^*\mathrm{d}f\\&=\int_{-\infty}^{\infty}P_X(f)\left[1-\mathrm{sgn}(f)\right]\mathrm{e}^{\mathrm{j}2\pi f\tau}\mathrm{e}^{-\mathrm{j}2\pi fu}\mathrm{d}f\end{aligned}$$

代入式(5)：

$$\begin{aligned}\mathrm{E}\left[Z_1(t+\tau)Z_2^*(t)\right]&=\int_{-\infty}^{\infty}h_1(u)\left\{\int_{-\infty}^{\infty}P_X(f)\left[1-\mathrm{sgn}(f)\right]\mathrm{e}^{\mathrm{j}2\pi f\tau}\mathrm{e}^{-\mathrm{j}2\pi fu}\mathrm{d}f\right\}\mathrm{d}u\\&=\int_{-\infty}^{\infty}P_X(f)\left[1-\mathrm{sgn}(f)\right]\mathrm{e}^{\mathrm{j}2\pi f\tau}\left\{\int_{-\infty}^{\infty}h_1(u)\mathrm{e}^{-\mathrm{j}2\pi fu}\mathrm{d}u\right\}\mathrm{d}f\end{aligned}\tag{6}$$

花括号部分是$h_1(t)$的傅氏变换，于是

$$\mathrm{E}\left[Z_1(t+\tau)Z_2^*(t)\right]=\int_{-\infty}^{\infty}P_X(f)\left[1-\mathrm{sgn}(f)\right]\mathrm{e}^{\mathrm{j}2\pi f\tau}\left[1+\mathrm{sgn}(f)\right]\mathrm{d}f\tag{7}$$

注意$[1+\mathrm{sgn}(f)][1-\mathrm{sgn}(f)]=0$，于是

$$\mathrm{E}\left[Z_1(t+\tau)Z_2^*(t)\right]=0\tag{8}$$

例 3 求下列复随机过程的均值、自相关函数、共轭相关函数。θ在$[0,2\pi]$内均匀分布。

(1) $Z(t)=\cos(2\pi t+\theta)+\mathrm{j}\cdot\sin(2\pi t+\theta)$

(2) $Z(t)=\cos(2\pi t+\theta)+2\mathrm{j}\cdot\sin(2\pi t+\theta)$;

解：(1) $Z(t)=\cos(2\pi t+\theta)+\mathrm{j}\cdot\sin(2\pi t+\theta)=\mathrm{e}^{\mathrm{j}(2\pi t+\theta)}$，其均值是

$$\mathrm{E}\left[Z(t)\right]=\mathrm{E}\left[\mathrm{e}^{\mathrm{j}2\pi t}\cdot\mathrm{e}^{\mathrm{j}\theta}\right]=\mathrm{e}^{\mathrm{j}2\pi t}\cdot\mathrm{E}\left[\mathrm{e}^{\mathrm{j}\theta}\right]=0$$

自相关函数是

$$\mathrm{E}\left[Z(t+\tau)Z^{*}(t)\right]=\mathrm{E}\left[\mathrm{e}^{\mathrm{j}2\pi(t+\tau)}\cdot\mathrm{e}^{\mathrm{j}\theta}\times\left\{\mathrm{e}^{\mathrm{j}2\pi t}\cdot\mathrm{e}^{\mathrm{j}\theta}\right\}^{*}\right]=\mathrm{E}\left[\mathrm{e}^{\mathrm{j}2\pi\tau}\right]=\mathrm{e}^{\mathrm{j}2\pi\tau}$$

共轭相关函数是

$$\mathrm{E}\left[Z(t+\tau)Z(t)\right]=\mathrm{E}\left[\mathrm{e}^{\mathrm{j}2\pi(t+\tau)}\cdot\mathrm{e}^{\mathrm{j}\theta}\times\left\{\mathrm{e}^{\mathrm{j}2\pi t}\cdot\mathrm{e}^{\mathrm{j}\theta}\right\}\right]=\mathrm{E}\left[\mathrm{e}^{\mathrm{j}(4\pi t+2\pi\tau)}\cdot\mathrm{e}^{\mathrm{j}2\theta}\right]=0$$

(2) 可将本小题中的$Z(t)$表示为

$$\begin{aligned}Z(t)&=\cos(2\pi t+\theta)+2\mathrm{j}\cdot\sin(2\pi t+\theta)\\&=\frac{\mathrm{e}^{\mathrm{j}(2\pi t+\theta)}+\mathrm{e}^{-\mathrm{j}(2\pi t+\theta)}}{2}+2\mathrm{j}\cdot\frac{\mathrm{e}^{\mathrm{j}(2\pi t+\theta)}-\mathrm{e}^{-\mathrm{j}(2\pi t+\theta)}}{2\mathrm{j}}=\frac{3}{2}\mathrm{e}^{\mathrm{j}(2\pi t+\theta)}-\frac{1}{2}\mathrm{e}^{-\mathrm{j}(2\pi t+\theta)}\end{aligned}$$

其均值为

$$\mathrm{E}\left[Z(t)\right]=\mathrm{E}\left[\frac{3}{2}\mathrm{e}^{\mathrm{j}(2\pi t+\theta)}-\frac{1}{2}\mathrm{e}^{-\mathrm{j}(2\pi t+\theta)}\right]=0$$

自相关函数是

$$\begin{aligned}\mathrm{E}\left[Z(t+\tau)Z^{*}(t)\right]&=\mathrm{E}\left[\left\{\frac{3}{2}\mathrm{e}^{\mathrm{j}(2\pi t+2\pi\tau+\theta)}-\frac{1}{2}\mathrm{e}^{-\mathrm{j}(2\pi t+2\pi\tau+\theta)}\right\}\left\{\frac{3}{2}\mathrm{e}^{-\mathrm{j}(2\pi t+\theta)}-\frac{1}{2}\mathrm{e}^{\mathrm{j}(2\pi t+\theta)}\right\}\right]\\&=\mathrm{E}\left[\frac{9}{4}\mathrm{e}^{\mathrm{j}2\pi\tau}+\frac{1}{4}\mathrm{e}^{-\mathrm{j}2\pi\tau}-\frac{3}{4}\mathrm{e}^{\mathrm{j}(4\pi t+2\pi\tau+2\theta)}-\frac{3}{4}\mathrm{e}^{-\mathrm{j}(4\pi t+2\pi\tau+2\theta)}\right]\\&=\frac{9}{4}\mathrm{e}^{\mathrm{j}2\pi\tau}+\frac{1}{4}\mathrm{e}^{-\mathrm{j}2\pi\tau}\end{aligned}$$

共轭相关函数是

$$\begin{aligned}\mathrm{E}\left[Z(t+\tau)Z(t)\right]&=\mathrm{E}\left[\left\{\frac{3}{2}\mathrm{e}^{\mathrm{j}(2\pi t+2\pi\tau+\theta)}-\frac{1}{2}\mathrm{e}^{-\mathrm{j}(2\pi t+2\pi\tau+\theta)}\right\}\left\{\frac{3}{2}\mathrm{e}^{\mathrm{j}(2\pi t+\theta)}-\frac{1}{2}\mathrm{e}^{-\mathrm{j}(2\pi t+\theta)}\right\}\right]\\&=\mathrm{E}\left[\frac{9}{4}\mathrm{e}^{\mathrm{j}(4\pi t+2\pi\tau+2\theta)}+\frac{1}{4}\mathrm{e}^{-\mathrm{j}(4\pi t+2\pi\tau+2\theta)}-\frac{3}{4}\mathrm{e}^{\mathrm{j}2\pi\tau}-\frac{3}{4}\mathrm{e}^{-\mathrm{j}2\pi\tau}\right]\\&=-\frac{3}{4}\mathrm{e}^{\mathrm{j}2\pi\tau}-\frac{3}{4}\mathrm{e}^{-\mathrm{j}2\pi\tau}=-\frac{3}{2}\cos 2\pi\tau\end{aligned}$$